BIOCHEMISTRY CUMULATIVE EXAM

January 26, 2001

1. (60 points) The amino acid compositions of a number of proteins are given below, with the subscripts indicating the number of amino acids of each type in the protein. For each protein, estimate the net charge at pH 4, pH 7, and pH 10, as well as the isoelectric pI. You may assume the pK_a values given in the table. Show your work.

a. $Ala_{10}Leu_{15}Arg_{10}Ser_{10}Trp_{10}$

b. $Ala_{10}Leu_{15}Arg_{5}Glu_{5}Ser_{10}Trp_{10}$

c. $Ala_{10}Leu_{15}Arg_{5}His_{5}Ser_{10}Trp_{10}$

d. $Ala_{10}Leu_{15}Glu_{5}His_{5}Ser_{10}Trp_{10}$

e. $Ala_{10}Leu_{15}Glu_{10}Ser_{10}Trp_{10}$

f. $Ala_{10}Leu_{15}His_{10}Ser_{10}Trp_{10}$

2. (40 points) Propose a practical large-scale method to separate the above proteins from a mixture containing approximately 100 mg of each protein. Describe the method in detail, giving the materials and supplies, specific conditions of the separation, and length of time for the procedure. Describe the principles on which the separation is based.

Table 5-4 **pK′ Values for the Ionizing Groups of Some Amino Acids at 25°C**

Amino acid	pK_1' —COOH	pK_2' $—NH_3^+$	pK_R' R group
Glycine	2.34	9.6	
Alanine	2.34	9.69	
Leucine	2.36	9.60	
Serine	2.21	9.15	
Threonine	2.63	10.43	
Glutamine	2.17	9.13	
Aspartic acid	2.09	9.82	3.86
Glutamic acid	2.19	9.67	4.25
Histidine	1.82	9.17	6.0
Cysteine	1.71	10.78	8.33
Tyrosine	2.20	9.11	10.07
Lysine	2.18	8.95	10.53
Arginine	2.17	9.04	12.48